

## Problem

|   |   |              |
|---|---|--------------|
| 0 | 1 | [0, 1, 0, 0] |
| 1 | 1 | [0, 2, 0, 0] |
| 2 | 3 | [0, 2, 0, 1] |
| 3 | 2 | [0, 2, 1, 1] |
| 4 | 1 | [0, 3, 1, 1] |

## Submissions

The frequency array is **[0, 3, 1, 1]**. These values can be used to create the sorted array as well: **sorted = [1, 1, 1, 2, 3]**.

## Note

For this exercise, always return a frequency array with 100 elements. The example above shows only the first 4 elements, the remainder being zeros.

## Leaderboard

## Challenge

Given a list of integers, count and return the number of times each value appears as an array of integers.

## Function Description

Complete the countingSort function in the editor below.

countingSort has the following parameter(s):

- arr[n]: an array of integers

## Returns

- int[100]: a frequency array

## Input Format

The first line contains an integer **n**, the number of items in **arr**.

Each of the next **n** lines contains an integer **arr[i]** where  $0 \leq i < n$ .

## Constraints

$$100 \leq n \leq 10^6$$

$$0 \leq arr[i] < 100$$

## Sample Input

```
100
63 25 73 1 98 73 56 84 86 57 16 83 8 25 81 5
```

## Sample Output

```
0 2 0 2 0 0 1 0 1 2 1 0 1 1 0 0 2 0 1 0 1 2
```

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Language Python 3



```
1 def countingSort(arr):
2     count = [0] * 100
3
4     for i in arr:
5         count[i] += 1
6
7     return count
8
9
10 n = int(input())
11 arr = list(map(int, input().split()))
12
13 result = countingSort(arr)
14 print(*result)
15
```

Line: 14 Col: 15

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☐ Test against custom input

## Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Compiler Message

Test case 1

Success

Test case 2

Test case 3

Test case 4

Test case 5